

# Formalizing Copyright Law through melody similarity

An explainable decision-support prototype for music copyright disputes

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**Project claim.** We do not automate copyright adjudication. We make one legally relevant issue explicit and inspectable: whether two songs share a recognizable melodic fragment.

## 1 Legal Problem

Music copyright disputes often turn on the open-textured standard of **substantial similarity**. But similarity is not itself a legal conclusion.

- Courts also consider access, protectability, scope of the disputed material, and context.
- Raw similarity scores can be persuasive but legally misleading if their meaning is hidden.
- The research gap is not only computational; it is techno-legal: how can a score support legal reasoning without pretending to replace it?

**Research question.** Can local melodic alignment capture legally relevant fragment similarity better than global comparison while remaining explainable enough for decision support?

## 2 Achieved Scope

The fall pitch proposed a broad formalization pipeline. The poster-stage prototype focuses on the part that can be defended with the current code and data.

| Initial ambition    | Achieved component                                                                 |
|---------------------|------------------------------------------------------------------------------------|
| Legal formalization | Minimal legal variables and rule-trace logic.                                      |
| Similarity layer    | Needleman–Wunsch global alignment, Smith–Waterman local alignment, and PMI scores. |
| Threshold analysis  | Exploratory thresholds on 15 melody-dispute pairs.                                 |
| Decision support    | Scorecard interpretation, not automated judgment.                                  |

**Why this is credible.** It presents the working evidence layer clearly: what it measures, where it helps, and where legal judgment remains in control.

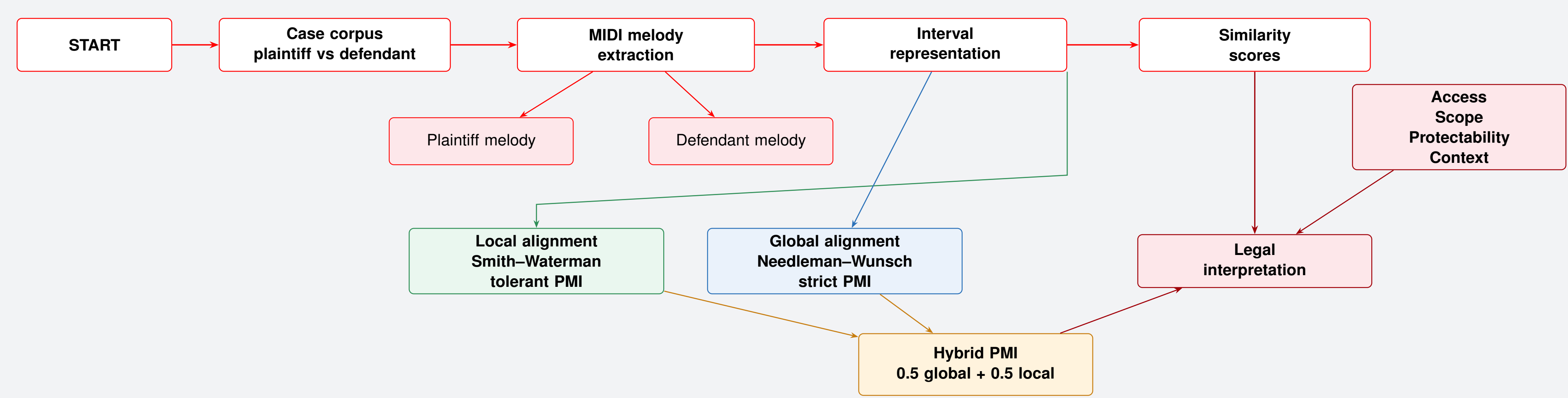
## 3 Dataset and Legal Coding

| Artifact          | Current state                                           |
|-------------------|---------------------------------------------------------|
| Corpus            | 128 music copyright disputes in the master case file.   |
| Evaluation subset | 15 pairs with curated MIDI melody excerpts.             |
| Representation    | Monophonic pitch sequences converted to intervals.      |
| Legal variables   | Access, melody scope, protectability, context override. |

**Bounded contribution.** The project is strongest when presented as an explainable similarity and legal-interpretation layer for melody-focused cases.

## 4 Method Overview

**Implemented pipeline.** MIDI melody excerpts are transformed into interval sequences, aligned globally and locally, then interpreted through a minimal legal rule layer.



**Melody representation.**

$$I = [p_2 - p_1, p_3 - p_2, \dots, p_n - p_{n-1}]$$

Intervals make comparison key-invariant and focused on melodic contour.

**Alignment layer.**

|                                                                                                         |                                                                                                              |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Global</b><br>Aligns the whole interval sequence and penalizes gaps; useful for overall resemblance. | <b>Local</b><br>Finds the best matching subsequence; useful when alleged copying is a recognizable fragment. |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|

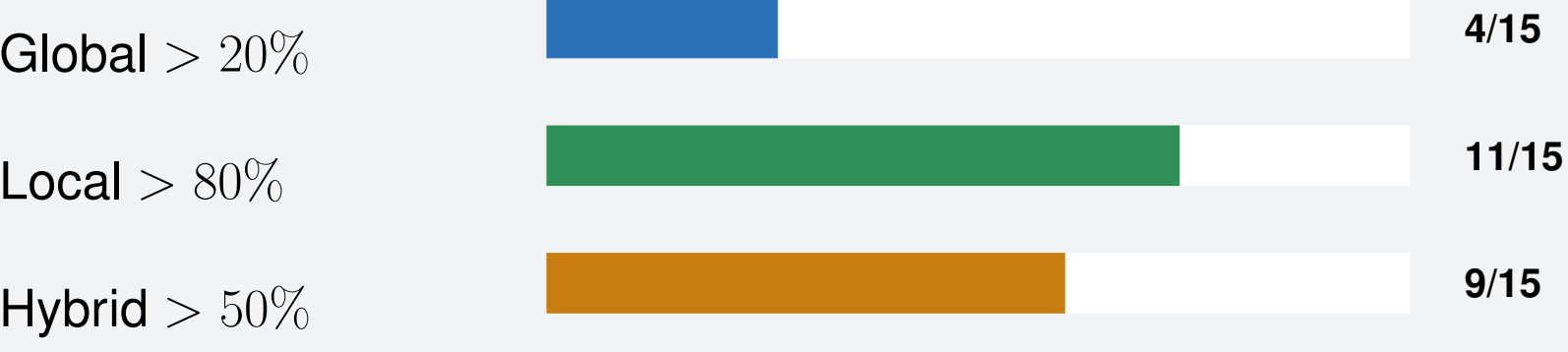
**Global PMI**  
Whole-excerpt resemblance. Conservative, but rigid when copying is fragment-based.

**Local PMI**  
Best-matching fragment resemblance. Closer to legally relevant motif copying.

**Hybrid PMI**  
Balances whole-melody and fragment evidence so one local match does not decide the case alone.

## 5 Results

**Threshold behavior on 15 curated melody pairs.**



**Main empirical finding.** Local PMI flags more of the 15 curated melody-dispute pairs under the exploratory threshold than global PMI.

**Legal caution.** High similarity is over-inclusive: a score needs access, protectability, scope, and context before it becomes legally meaningful.

| Illustrative readings.                |       |       |
|---------------------------------------|-------|-------|
| Pair                                  | G     | L     |
| No Scrubs / Shape of You              | 8.39  | 92.12 |
| Bitter Sweet Symphony / The Last Time | 26.95 | 77.54 |
| Ice Ice Baby / Under Pressure         | 2.78  | 33.33 |
| Flowers / When I Was Your Man         | 36.10 | 89.43 |

**Full score table.** Global and local scores are shown separately; the hybrid score is their simple average.

| #  | Melody pair                                      | Global | Local | Hybrid |
|----|--------------------------------------------------|--------|-------|--------|
| 1  | The Last Time / Bitter Sweet Symphony            | 26.95  | 77.54 | 52.24  |
| 2  | Anybody Seen My Baby / Constant Craving          | 5.01   | 93.08 | 49.04  |
| 3  | All You Need Is Love / In the Mood               | 12.85  | 95.05 | 53.95  |
| 4  | Oops Up Side Your Head / Uptown Funk             | 6.58   | 62.90 | 34.74  |
| 5  | Greatest Love of All / If You Could Read My Mind | 21.10  | 93.23 | 57.16  |
| 6  | Sweet Little Sixteen / Surfin' U.S.A.            | 18.62  | 93.70 | 56.16  |
| 7  | I Won't Back Down / Stay With Me                 | 7.76   | 88.89 | 48.32  |
| 8  | No Scrubs / Shape of You                         | 8.39   | 92.12 | 50.25  |
| 9  | I'm Not Alone / Yeah 3x                          | 3.61   | 0.00  | 1.81   |
| 10 | Teach the World to Sing / Shakermaker            | 12.90  | 89.46 | 51.18  |
| 11 | Under Pressure / Ice Ice Baby                    | 2.78   | 33.33 | 18.06  |
| 12 | Every Breath You Take / I'll Be Missing You      | 23.01  | 93.35 | 58.18  |
| 13 | All Day and All of the Night / Hello, I Love You | 12.74  | 93.94 | 53.34  |
| 14 | I Want a New Drug / Ghostbusters                 | 9.91   | 85.00 | 47.45  |
| 15 | When I Was Your Man / Flowers                    | 36.10  | 89.43 | 62.77  |

**Interpretation.** Global PMI is useful as a conservative baseline; local PMI is better aligned with the legal intuition that a recognizable fragment can matter even when the full melodies differ.

**Global result.** Conservative whole-excerpt comparison flags only 4/15 pairs, so it misses several fragment-centered disputes.

**Local result.** Best-fragment comparison flags 11/15 pairs and better matches the way melody disputes are argued.

**Hybrid result.** The combined score flags 9/15 pairs, keeping sensitivity while avoiding reliance on one fragment alone.

## 6 Legal Interpretation

The contribution is not that scores decide cases. The contribution is a transparent structure for interpreting scores.

| Factor         | Role                                                                  |
|----------------|-----------------------------------------------------------------------|
| Access         | Similarity is weaker without plausible access.                        |
| Melody scope   | The tool should not decide lyrics, timbre, production, or vibe cases. |
| Protectability | High similarity can still concern generic material.                   |
| Context        | Unusual cases require manual legal review.                            |

**Rule-trace logic.** Stop if outside melodic scope; lower concern if access is absent; flag review only when similarity and legal factors align.

## 7 Discussion

- Empirical:** local alignment better captures recognizable fragments than full-excerpt comparison.
- Legal:** similarity is evidence, not the legal test itself.
- Computational:** the model is interpretable because each score has a clear methodological meaning.
- Capstone value:** the work connects a measurable artifact to a legal reasoning structure.

**Takeaway.** Computation can clarify one part of copyright reasoning, but it should support human legal judgment rather than replace it.

## 8 Conclusion

**Final claim.** The project achieved a defensible part of the original capstone goal: an explainable computational layer for melody-based copyright analysis.

| Limitation               | Next step                                  |
|--------------------------|--------------------------------------------|
| Melody only              | Add rhythm, harmony, sampling, production. |
| 15 MIDI pairs            | Expand and balance evaluation.             |
| Settlements as labels    | Separate settlements from court findings.  |
| Exploratory thresholds   | Calibrate and report uncertainty.          |
| Legal coding exploratory | Validate annotations and rule traces.      |

**Final takeaway.** The prototype turns melodic similarity from a black-box number into auditable evidence: useful for legal reasoning, but never a substitute for legal judgment.